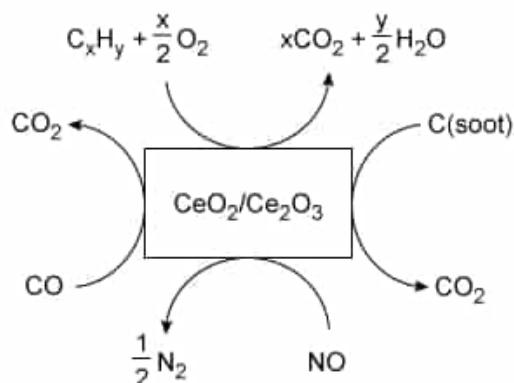


The diagram below shows the activity of cerium oxides that have been proposed as catalytic materials to reduce emissions of nitrogen monoxide, carbon monoxide, soot, and hydrocarbons during the combustion of diesel fuel. Based on the diagram, if the reactions are irreversible, which of the following does NOT illustrate a function of the cerium oxide catalyst?

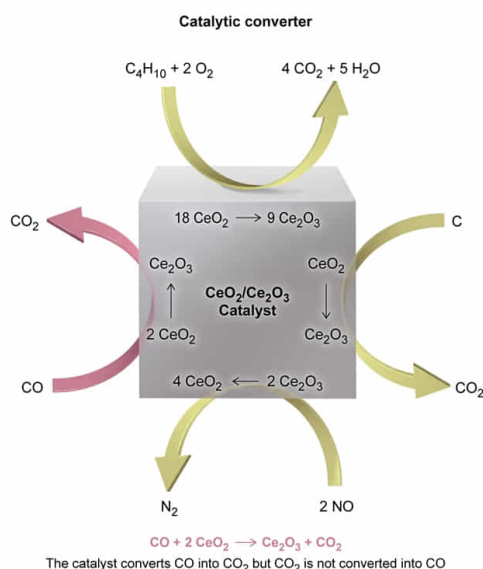


- ☐ A. $2 \text{ NO} + 2 \text{ Ce}_2\text{O}_3 \rightarrow 4 \text{ CeO}_2 + \text{N}_2$ [5%]
- ✓ ☒ B. $\text{CO}_2 + \text{Ce}_2\text{O}_3 \rightarrow 2 \text{ CeO}_2 + \text{CO}$ [79%]
- ☐ C. $\text{C}_4\text{H}_{10} + 2 \text{ O}_2 + 18 \text{ CeO}_2 \rightarrow 9 \text{ Ce}_2\text{O}_3 + 4 \text{ CO}_2 + 5 \text{ H}_2\text{O}$ [8%]
- ☐ D. $\text{C} + 4 \text{ CeO}_2 \rightarrow 2 \text{ Ce}_2\text{O}_3 + \text{CO}_2$ [6%]

Correct

80% answered correctly

Explanation:

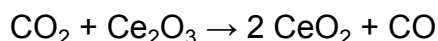


In **catalytic diagrams**, a **curved reaction arrow** depicts the conversion of one chemical species into another through the interaction with a **catalyst**. If several reactions are possible (or occur simultaneously), these conversions may be shown together. Because these diagrams focus primarily on the conversions that take place, some details such as **intermediate species** or regeneration of the catalyst may not be explicitly shown.

In the diagram given for the cerium oxide catalyst, the exact roles of the CeO_2 and Ce_2O_3

species are not given. Instead, the function of the catalyst is illustrated by showing four chemical conversions that the catalyst performs. Individual chemical reactions describing these chemical conversions should be in agreement with the diagram.

The diagram of the cerium oxide catalyst shows that CO is converted into CO₂; therefore, if the reactions are irreversible, the catalyst would NOT function for the *reverse reaction* (converting CO₂ to CO), which is depicted by the reaction



In the *reverse reaction*, CO₂ is a reactant and CO is a product, which is not shown in the diagram.

(Choices A and D) The diagram shows that the cerium oxides facilitate the conversion of NO into N₂ by a 2:1 molar ratio, as well as the oxidation of carbon soot to CO₂ in 1:1 molar proportions. These reactions are balanced and consistent with the diagram.

(Choice C) The diagram indicates that for a hydrocarbon comprised of x atoms of C and y atoms of H, the cerium oxides use x/2 atoms of O₂ to give an enhanced combustion reaction that produces x atoms of CO₂ and y/2 atoms of H₂O. This reaction for butane is balanced and is consistent with these proportions from the diagram.

Educational objective:

Catalytic diagrams use curved reaction arrows to depict the conversion of one chemical species into another through the interaction with a catalyst. These diagrams focus on the function of the catalyst and may not explicitly show intermediate species. Chemical reactions should be consistent with the catalyst function.

Time spent:0 sec

Subject:
General Chemistry

QID:402124

System:
Thermodynamics, Kinetics & Gas Laws